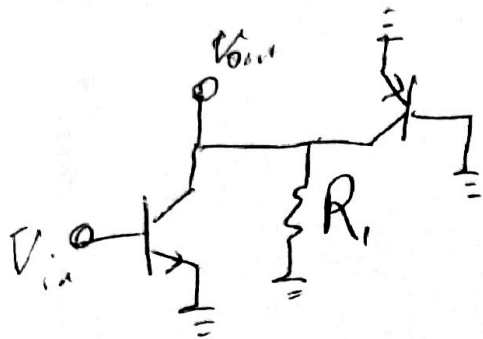


سوال 1 -

$$I_{C1} = I_{C2} = \frac{I_{EE}}{2} = 1 \text{ mA}$$

$$I_{C3} = I_{C4} \approx 1 \text{ mA} \quad : \text{در این مدار } \beta \text{ افتاد است}$$



$$g_m = \frac{I_C}{V_T} \quad : \text{تشدیدی}$$

$$r_o = \frac{V_A}{I_C}$$

$$A = -g_m r_{o1} \parallel r_{o3} \parallel R = 50$$

$$g_{m1} = 40 \text{ m}\Omega^{-1}$$

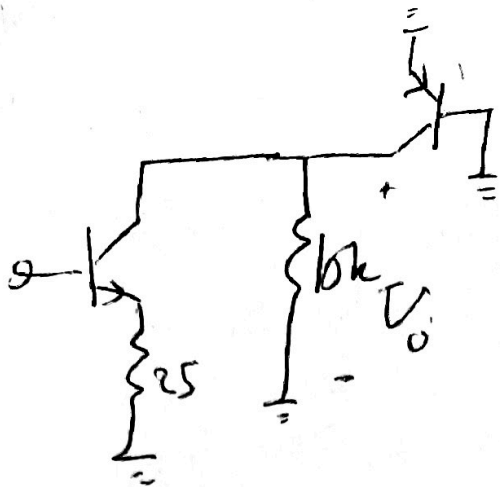
$$r_{o1} = 5 \text{ k}\Omega$$

$$r_{o3} = 4 \text{ k}\Omega$$

$$\left(+ \frac{g_{m1}}{50} - \frac{1}{r_{o1}} - \frac{1}{r_{o3}} \right)^{-1} = R = 2.86 \text{ k}\Omega$$

سوال (2)

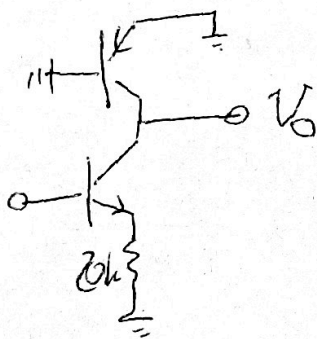
تفاضلی



$$A_v = - \frac{10k \parallel R_{o3} \parallel R_{o1}}{r_{m1} + 25}$$

$$R_{o1} = r_{o1} \frac{1 + g_m R_E}{1 + g_m R_{E/p}} \approx 2r_{o1}$$

$$A_v = -80$$



مكرر :-

$$A_{vcm} = - \frac{r_{o3} \parallel R_{o1}}{r_{m1} + 20k}, \quad R_{o1} = 89k\Omega$$

$$A_{vcm} \approx - \frac{r_{o3}}{r_{m1} + 20k} = 0.5$$

13د

$$I_{C1} = I_{C2} = 0.5 \text{ mA}$$

$$V_{CE1} = 5 - 3 - 6 = 1.4 \text{ V}$$

$$\rightarrow I_{C3} = I_{C4} = \frac{5.7}{2 \times 3.8 \text{ k}\Omega} = 0.75 \text{ mA}$$

$$g_{m1} = 20 \text{ m}\Omega^{-1}$$

$$r_{\pi 1} = 10 \text{ k}\Omega$$

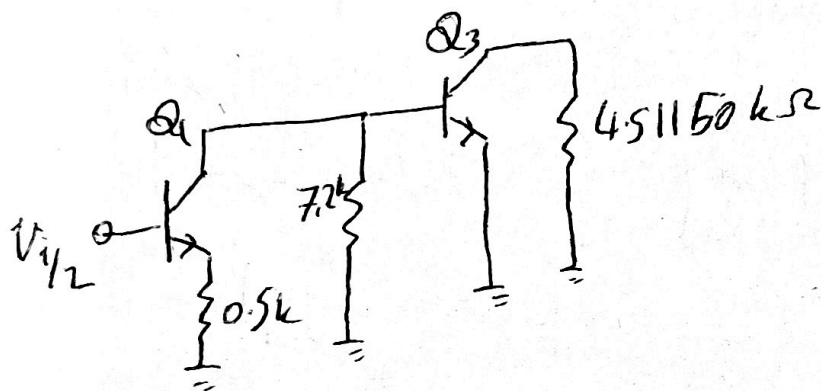
$$r_{o1} = 26 \text{ k}\Omega$$

$$g_{m2} = 30 \text{ m}\Omega^{-1}$$

$$r_{\pi 3} = 6.6 \text{ k}\Omega$$

$$r_{o3} = 173.3 \text{ k}\Omega$$

تفصیلی :

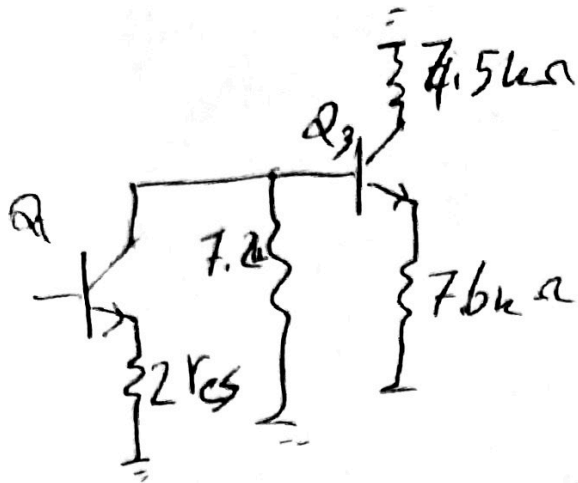


$$A_{Q3} = - \frac{r_{o3} \parallel 4.5 \text{ k}\Omega \parallel 50 \text{ k}\Omega}{r_{m3} + 0} =$$

$$A_{Q1} = - \frac{7.2 \text{ k}\Omega \parallel r_{\pi 3} \parallel R_{o1}}{r_{m1} + 0.5 \text{ k}\Omega}$$

$$\rightarrow A = A_{Q3} A_{Q1} = 748$$

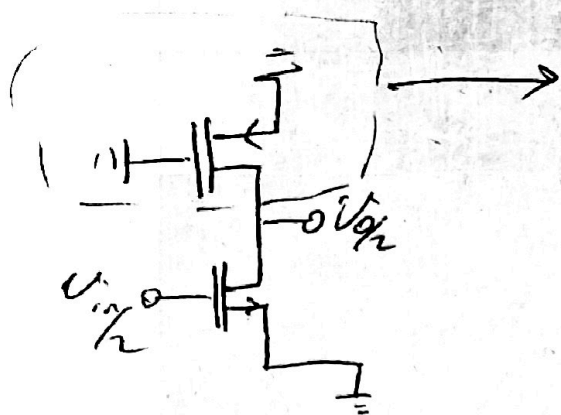
سوال 3 - ادا



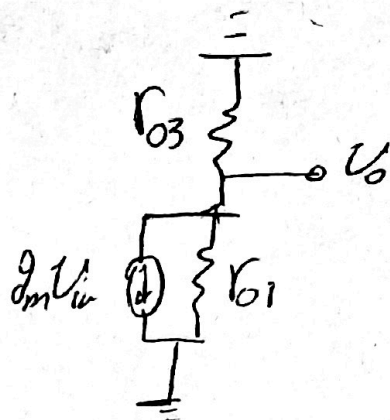
$$A_{Q3} = - \frac{4.5k \parallel R_{O3}}{r_{m3} + 7.6k\Omega}$$

$$A_{Q1} = - \frac{7.2k \parallel (r_{\pi3} + \beta(7.6k)) \parallel R_{O1}}{r_{m1} + 400k}$$

$$A_{cm} = 0.011$$



تفاضلی :



$$v_o = -g_m v_{in} (r_{o1} \parallel r_{o3}) \quad , \quad g_m = \frac{2 \bar{I}_D}{v_{oD}}$$

$$r_o = \frac{1}{\lambda \bar{I}_D} \quad r_{o1} = 2 r_{o3}$$

$$\bar{I}_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{oD})^2$$

$$P = 2 \bar{I}_{D3} V_{DD} \rightarrow \bar{I}_{D3} = 0.56 \text{ mA}$$

$$A_v = -g_m r_{o3} = -\frac{2 \bar{I}_D}{3 V_{oD}} \frac{1}{\lambda \bar{I}_D}$$

$$V_{oD} = 0.167 \text{ V}$$

$$\rightarrow (W/L)_1 = 398 \approx 400$$

$$(W/L)_2 \approx 800$$